higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION – SHEET METAL WORK
NQF LEVEL 4

(6031024)

22 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only blue or black ink.
5. Write neatly and legibly.

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QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 A spiral duct is a round duct made from a continuous strip of sheet metal.

1.2 CNC machines are not computer controlled.

1.3 Fans are used to move air around.

1.4 Acetylene gas cylinders can be stored in a horizontal position.

1.5 A transition piece is a duct fitting that changes the size of a duct without changing the direction.

(5 × 1) **[5]**

QUESTION 2

Choose a description from COLUMN B that matches a/an word/item in COLUMN A. Write only the letter (A–J) next to the question number (2.1–2.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1	Heel radius	A	protect the welder's eyes from ultra-violet rays
2.2	Spatters can be prevented by	B	iron contaminants have come into contact with the stainless steel
2.3	Quadrant	C	computer processing unit
2.4	Software	D	set of instructions used for solving mathematical problems
2.5	Galvanized steel	E	radius of the arc that forms the elbow outside radius/heel
2.6	Helmet	F	area created between the horizontal and vertical axis
2.7	A post-processor	G	conducts heat very well
2.8	Pickling and passivation is done on stainless steel	H	shortening the length of the arc
2.9	Algorithm	I	does not rust or corrode easily
2.10	Non-ferrous metals	J	program that tells the computer how to do a particular task

(10 × 1) **[10]**

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QUESTION 3

3.1 State TWO differences between *flex ducts* and *spiral ducts* in relation to their length. (2 × 2) (4)

3.2 Explain what is meant by the term *thermal acquisition*. (2)

3.3 State FIVE pieces of information needed before a draughtsman can prepare a plant layout drawing. (5)

3.4 Describe the following terms regarding air diffusion terminals within a ductwork layout:

3.4.1 Forced ventilation

3.4.2 Infiltration

3.4.3 Natural ventilation

(3 × 2) (6)
[17]

QUESTION 4

4.1 Define the following terms that deal with air movement:

4.1.1 Condensate draining

4.1.2 Mixing chamber

(2 × 2) (4)

4.2 Define the following terms used for rectangular and square duct elbows:

4.2.1 Cheek

4.2.2 Heel

4.2.3 Throat radius

(3 × 2) (6)

4.3 List FIVE main pre-operational checks/tests that must be carried out on TIG welding equipment. (5)

4.4 State the line thickness according to the drawing convention, for drawing a schematic layout of an air conditioning plant. (1)
[16]

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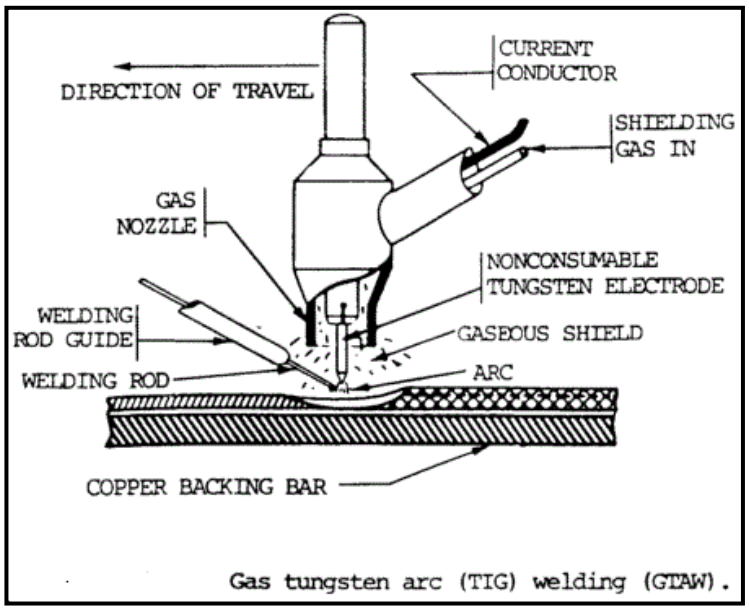
QUESTION 5

5.1 5.1.1 Name THREE main possible shielding gases that one would use when operating a TIG welding machine. (3)

5.1.2 Which gas in QUESTION 5.1.1 would allow the torch to travel faster and increase penetration? (1)

5.1.3 What is the colour code at the tip of the tungsten electrode for welding stainless steel by means of the TIG welding process? (1)

5.2 Study the diagram below and answer QUESTIONS 5.2.1–5.2.3.



5.2.1 What is the direction of travel indicated by the arrow? (1)

5.2.2 The gas nozzle is made of a non-ferrous metal.

What could be the reason? (2)

5.2.3 Why is the welding electrode classified as a non-consumable electrode? (2)

5.3 Explain the term *hydrogen cracking*. (2)

5.4 State FIVE disadvantages of TIG welding. (5)

[17]

QUESTION 6

6.1 Identify the THREE linear axes and indicators used in CNC lathes with reference to the right-hand rule. (3)

6.2 During which THREE situations should the reference point 'R' (Starting point) be used? (3)

6.3 Name FOUR components of a CNC lathe that are identical to those of a manual machine. (4)

6.4 Briefly explain the JIT principle. (3)

6.5 Give a brief description and ONE example of the purpose of CNC machines related to the following industries.

6.5.1 Engineering and machining industry

6.5.2 Fabrication industry

6.5.3 Woodworking industry

(3 × 2) (6)

6.6 State the THREE main steps that a CNC operator must follow when preparing the machine for use. (3)

6.7 As a regular CNC machine operator, which THREE storage devices can be used to back-up the program stored on the CNC computer? (3)

6.8 Write a brief explanation about the cleaning of a CNC machine after use. Your explanation must be based on the following phrases:

6.8.1 Removal of waste materials

6.8.2 Separate waste

6.8.3 Company's procedure on waste

(3 × 2) (6)

6.9 State the TWO conditions why tool testing must be allowed before and after use. (2 × 2) (4)

[35]

TOTAL: 100

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